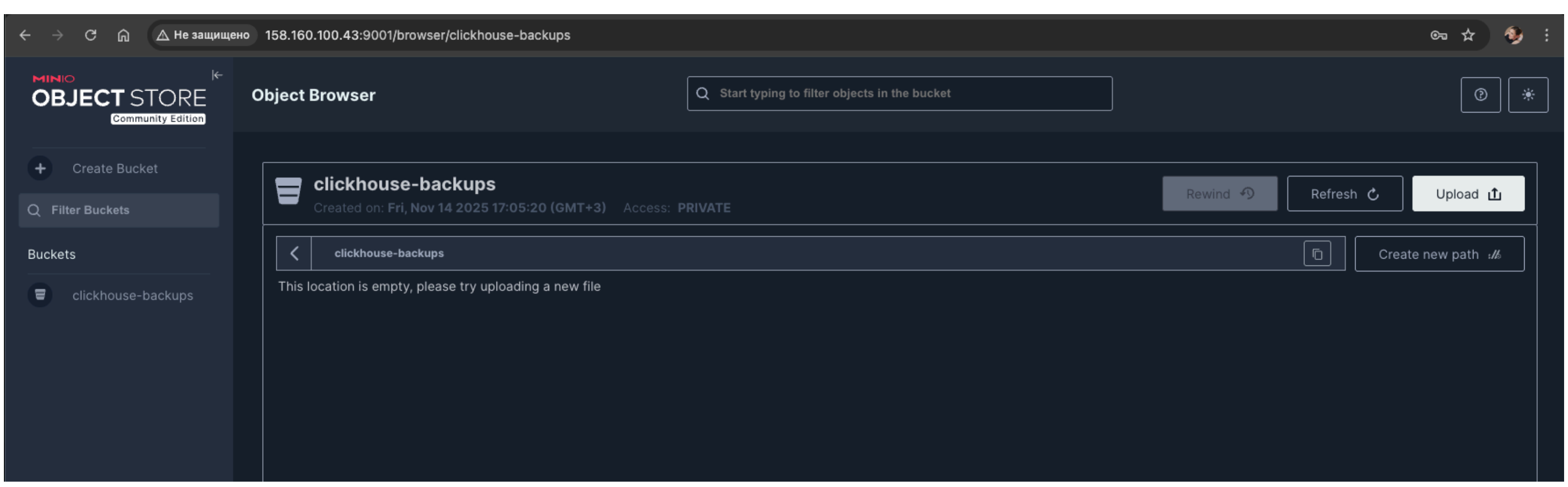
StoragePolicy

1. Развернуто minIO

```
 yc compute instance create --name minio-node --ssh-key /Users/zhannamakrusina/yandex-cloud/
key_ssh_id.pub --create-boot-disk image-folder-id=standard-images,image-family=ubuntu-2404-
lts,size=50,auto-delete=true --network-interface subnet-name=default-ru-central1-a,nat-ip-
version=ipv4 --memory 16G --cores 4 --core-fraction 50 --zone ru-central1-a --preemptible --
hostname minio-node

done (42s)
id: fhmpnrjkq9tom5dplt52
folder_id: b1g8hdofsv85pjmtq38l
created_at: "2025-11-14T13:48:31Z"
name: minio-node
zone_id: ru-central1-a
platform_id: standard-v2
resources:
  memory: "17179869184"
  cores: "4"
  core_fraction: "50"
status: RUNNING
metadata_options:
  gce_http_endpoint: ENABLED
  aws_v1_http_endpoint: ENABLED
  gce_http_token: ENABLED
  aws_v1_http_token: DISABLED
boot_disk:
  mode: READ_WRITE
  device_name: fhmhokme0hijd1qk2ku3
  auto_delete: true
  disk_id: fhmhokme0hijd1qk2ku3
network_interfaces:
  - index: "0"
    mac_address: d0:0d:19:be:e7:4d
    subnet_id: e9b7lj0aeh088c2g24af
    primary_v4_address:
      address: 10.128.0.3
      one_to_one_nat:
        address: 158.160.100.43
        ip_version: IPV4
serial_port_settings:
  ssh_authorization: OS_LOGIN
gpu_settings: {}
fqdn: minio-node.ru-central1.internal
scheduling_policy:
  preemptible: true
network_settings:
  type: STANDARD
placement_policy: {}
hardware_generation:
  legacy_features:
    pci_topology: PCI_TOPOLOGY_V1

There is a new yc version '0.175.0' available. Current version: '0.154.0'.
See release notes at https://yandex.cloud/ru/docs/cli/release-notes
You can install it by running the following command in your shell:
    $ yc components update


yc-user@minio-node:~$ wget https://dl.min.io/server/minio/release/linux-amd64/minio
```

```
--2025-11-14 13:51:00--  https://dl.min.io/server/minio/release/linux-amd64/minio
Resolving dl.min.io (dl.min.io)... 178.128.69.202, 138.68.11.125
Connecting to dl.min.io (dl.min.io)|178.128.69.202|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 110989496 (106M) [application/octet-stream]
Saving to: ‘minio’

minio               100%[===================>] 105.85M  17.9MB/s    in 7.2s

2025-11-14 13:51:08 (14.6 MB/s) - ‘minio’ saved [110989496/110989496]

yc-user@minio-node:~$ chmod +x minio
yc-user@minio-node:~$ sudo mv minio /usr/local/bin/
yc-user@minio-node:~$ sudo useradd -r minio-user

yc-user@minio-node:~$ export MINIO_ROOT_USER=minioadmin
yc-user@minio-node:~$ export MINIO_ROOT_PASSWORD=minioadmin
yc-user@minio-node:~$ sudo ufw allow 9000/tcp
Rules updated
Rules updated (v6)
[1]+  Exit 1                  nohup minio server /data/minio --address ":9000" --console-address
":9001" > minio.log 2>&1
yc-user@minio-node:~$ sudo ufw allow 9001/tcp
Rules updated
Rules updated (v6)

yc-user@minio-node:~$ mkdir -p ~/minio-data
yc-user@minio-node:~$ nohup minio server ~/minio-data --address ":9000" --console-address ":9001" >
~/minio.log 2>&1 &
[1] 1353
yc-user@minio-node:~$
```



Заходим на интерфейс и создаем bucket ckickhouse_backups

```
yc-user@clickhouse-node:/etc/clickhouse-server/config.d$ vim s3_storage.xml

<clickhouse>
    <storage_configuration>
        <disks>
            <minio_s3>
                <type>s3</type>
                <endpoint>http://158.160.100.43:9000/clickhouse-backups/</endpoint>
                <access_key_id>minioadmin</access_key_id>
                <secret_access_key>minioadmin</secret_access_key>
            </minio_s3>
        </disks>
        <policies>
            <s3_main>
                <volumes>
                    <main>
                        <disk>default</disk>
```

```
                    </main>
                    <external>
                        <disk>minio_s3</disk>
                    </external>
                </volumes>
            </s3_main>
        </policies>
    </storage_configuration>
</clickhouse>
```

Ставим clickhouse-backup

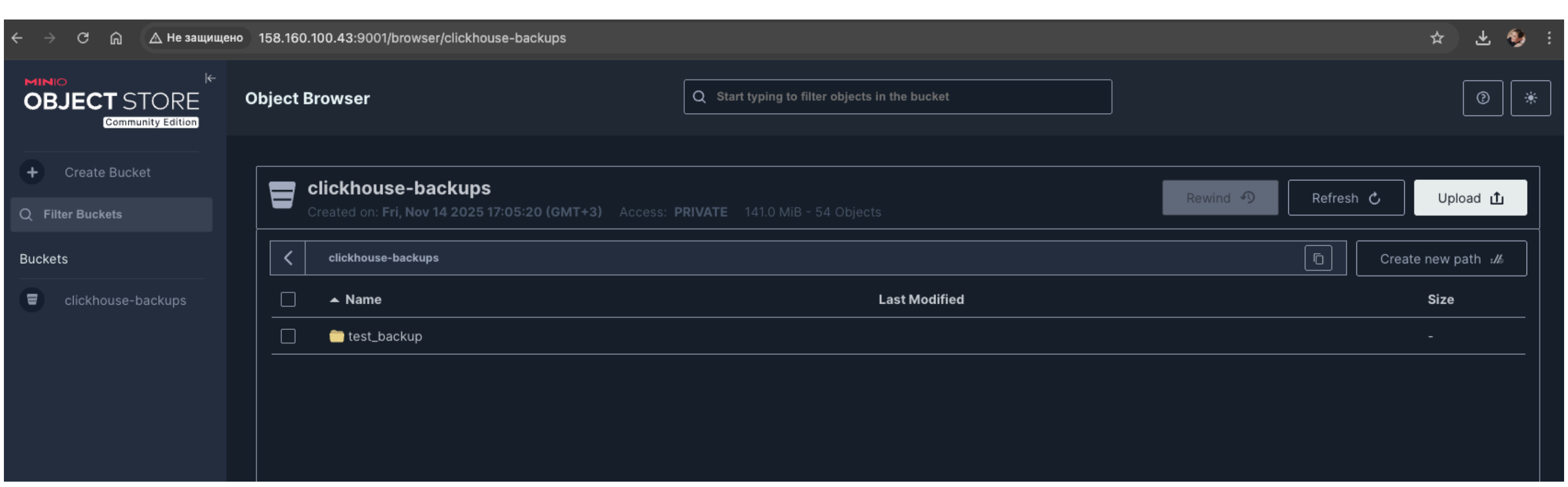
```
wget https://github.com/Altinity/clickhouse-backup/releases/latest/download/clickhouse-backup-linux-amd64.tar.gz
tar -xvf clickhouse-backup-linux-amd64.tar.gz
sudo mv build/linux/amd64/clickhouse-backup /usr/local/bin/
```

После установки выгружаем конфиг по умолчанию и правим его

```
clickhouse-backup default-config > config.yml
root@clickhouse-node:/etc/clickhouse-backup# vim config.yml
general:
    remote_storage: s3

s3:
    access_key: "minioadmin"
    secret_key: "minioadmin"
    bucket: "clickhouse-backups"
    endpoint: "http://158.160.100.43:9000"
    region: "us-east-1"
    acl: private
    assume_role_arn: ""
    force_path_style: true
    path: ""
    object_disk_path: ""
    disable_ssl: true
    compression_level: 1
    compression_format: tar
    sse: ""
    sse_kms_key_id: ""
    sse_customer_algorithm: «"
```

Создаем бекап и выгружаем его

```
clickhouse-backup create test_backup
clickhouse-backup upload test_backup
```



Удаляем таблицу:

```
 DROP TABLE test_db.users;

DROP TABLE test_db.users

Query id: b888f19c-f435-4b65-9ec2-5bcf1ebe70a4

Ok.
```

```
0 rows in set. Elapsed: 0.001 sec.

clickhouse-node.ru-central1.internal :) select * from test_db.users

SELECT *
FROM test_db.users

Query id: 6317fc9b-4dff-47bc-95d6-2595549becf3


Elapsed: 0.003 sec.

Received exception from server (version 25.3.5):
Code: 60. DB::Exception: Received from localhost:9000. DB::Exception: Unknown table expression
identifier ‘test_db.use
```

Восстанавливаем бекап

```
clickhouse-backup download test_backup
clickhouse-backup restore test_backup

 select * from test_db.users

SELECT *
FROM test_db.users

Query id: 262171c2-c2ac-4b55-9e79-4e55dc7f69e3

   ┌─id─┬─name──┐
1. │  1 │ Alice │
2. │  2 │ Bob   │
   └────┴───────┘

2 rows in set. Elapsed: 0.002 sec.
```

Заголовок